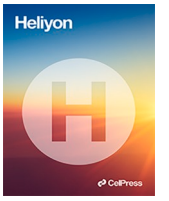
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Research article



Computational and experimental study on the aerodynamic performance of NACA 4412 airfoil with slot and groove

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ABSTRACT

Since the Wright Brothers' first flight in 1903, extensive research has been dedicated to improving the aerodynamic performance of aircraft. This study investigates the effect of two distinct wing geometric modifications on airfoil performance at high angles of attack (AOAs). These two modifications are slot, specifically the NACA 4412 with only a slot, and groove, specifically the NACA 4412 with both a slot and a groove. The investigation combines numerical simulation using ANSYS fluent with experimental evaluations conducted in the VDAS AF1300 subsonic wind tunnel. Since turbulent airflow often results in early stall, the primary objective of this research is to delay the stall angle of the normal NACA 4412 airfoil by mitigating local separation zones, and boundary layer transitions. Numerical simulations are performed at airspeeds of 20 m/s and 43.9 m/s, while experimental investigations are conducted at a speed of 20 m/s. The results indicate that both modified airfoils have higher lift-to-drag ratio than the normal airfoil at high AOAs. Specifically, the NACA 4412 airfoil with only a slot demonstrates the highest lift-to-drag ratio among the modified airfoils at high AOAs. Moreover, the NACA 4412 airfoil equipped with a slot and a groove demonstrates the highest stall angle, measured at 18°, compared to the normal NACA 4412 airfoil with a stall angle of 14°. At high AOA, the NACA 4412 airfoil with a slot generates a nearly 35 % higher lift coefficient than the normal NACA 4412 airfoil, while the NACA 4412 with a slot and a groove achieves almost a 16 % higher lift coefficient than the normal NACA 4412 airfoil.

1. Introduction

Enhancing airfoil efficiency stands as a widely explored area in engineering. There are many strategies, such as increasing lift, reducing drag, and delaying stall angle. But, understanding the boundary layer plays a pivotal role in understanding airfoil performance. Many researchers have studied various geometries, such as triangular rib [1], dimple [2], bumpy surface [3], and surface modifiers [4], to understand and control the flow separation. As at higher AOA, the flow may become unstable and separate from the wing, it causes a loss of lift and an increase in drag [5].

A novel custom-designed internal slot on an airfoil blade is proposed in a study by Ni, Z. et al. [6] as a passive flow control mechanism to improve lift and decrease drag. Slot width and inclined angles were adjusted for optimal design. A slotted NACA 63_4-021

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airfoil was compared with the baseline airfoil through simulation and experimental studies. The comparison was made in terms of different aerodynamic parameters, such as lift coefficients, drag coefficients, and lift-to-drag ratio at a Reynolds number of 10^5. The slotted airfoil showed higher lift and lower drag at angles of attack over 11°, but had a lower lift-to-drag ratio at smaller angles. Whitman, N. et al. [7] examined three airfoil configurations: a solid NACA 65_3-618 airfoil, and two slotted airfoils with small (0.25 in.) and large (0.375 in.) slots. The slots were aligned with the free stream flow at a 15° angle of attack, near the stagnation and separation points on the airfoil. The findings revealed that slotted airfoils outperformed the solid airfoil in lift generation at angles of attack above 15°, while underperforming slightly below this angle.

Numerous studies have been conducted on the use of slots within airfoils, with applications in helicopter, wind turbine and aircraft airfoil designs. For instance, Belamadi, R. et al. [8] focused on the effect of slot positioning and dimensions on the aerodynamic efficacy of wind turbine airfoils. Their findings showed that the presence of slots improve performance compared with the baseline configuration, especially at moderate to high angles of attack ranging from 10° to 20°. Beyhaghi, S. et al. [9] suggested that the incorporation of two-segment slots within the NACA 4412 airfoil resulted in a significant average increase of 8 % in lift coefficient across the entire range of angles of attack (AOAs). In a separate investigation, Beyhaghi, S. et al. [10] integrated leading-edge slots into the NACA 4412 airfoil with the goal of analyzing its aerodynamic characteristics. A significant increase in the lift coefficient of up to 30 % was observed in that study. Yeo, H. et al. [11] applied the forward-slotted airfoils to a UH-60A helicopter and then evaluated their performance. The findings revealed that these slotted airfoils had the potential to increase the maximum thrust output of the UH-60A helicopter by a remarkable 25 %, though a significant penalty was observed in their study.

Aşan, Ö. F. et al. [12] implemented various slotted structures at different chordwise locations on the NACA 0018 airfoil, conducting experiments at a Reynolds number of 3×10^5. Their investigation revealed that the slotted airfoil configurations generated increased lift forces, particularly notable at high angles of attack compared to the baseline airfoil. Furthermore, the slotted designs effectively delayed the stall angle. Chapman, J. et al. [13] conducted a parametric investigation aiming to identify the most effective slot configuration and discovered an optimal slot setup that significantly enhances the maximum lift coefficient by 27 %, eliminates separation points effectively, and postpones the stall angle by 2°. Fawzi, S. et al. [14] conducted a numerical investigation to determine the optimal slot position within the NACA 4412 airfoil, considering parameters such as slot exit location, width, and slope. Through this analysis, the authors identified the most effective slot design, characterized by an exit location at 60 % of the chord, a width equivalent to 1 % of the chord, and a slope of 65°. This configuration notably delays the stalling angle to 19° and enhances the maximum lift-to-drag ratio by 110 % at an angle of attack of 18°.

On the other hand, groove is also used to increase the aerodynamic performance. The concept of generating additional lift draws inspiration from golf balls. Golf ball with dimples can cover a greater distance compared to a smooth one. This is because the dimples generate increased momentum, which postpones the separation of flow and consequently reduces overall drag [15]. Through numerical simulations employing Reynolds-averaged Navier Stokes equations with modified wall boundary conditions, Wang, L. et al. [16] demonstrated that micro/nano grooved surfaces delay the transition from laminar to turbulent flow, resulting in a significant reduction in skin friction. This study also revealed a more stable near-wall flow field for grooved airfoils compared to smooth airfoils, with a notable decrease of 13.39 % in maximum turbulent intensity.

Szczepaniak, R. et al. [17] focused on grooved airfoils known as Kline-Fogleman airfoils, which are well-suited for low-load flights like private planes, drones, and RC vehicles due to their high stall angle. Despite not always producing higher lift, these airfoils showed stall resistance, enhancing aerodynamic performance at crucial attack angles, as verified by simulations. Venkatesh, D. T. et al. [18] used Computational Fluid Dynamics (CFD) to study the impact of adding grooves to a NACA 0018 airfoil. At different AOAs, lift and drag were measured. The 7 mm groove placed at 50 % of the chord provided the highest lift-to-drag ratio by reattaching the flow and increasing the stall angle to 16°, compared to the plain wing with a 14.5-degree stall angle. Almusawi, M. et al. [19] explored the impact of a spanwise semicircular groove on the aerodynamic performance of a NACA0012 airfoil through CFD simulations employing the k-ω turbulence model. The investigation involved comparing lift and drag coefficients between airfoils with and without the groove, while maintaining a consistent flow velocity of 20 m/s across all angles of attack. Results indicated that the inclusion of the semicircular groove enhanced the lift efficiency coefficient by 2.25 percent while concurrently reducing the drag coefficient by 4.32 percent.

Passive flow control techniques [20] for flight vehicles operating at low Reynolds numbers is another way to control the flow separation of an aircraft. Özden, M. et al. [21] examined the impacts of applying single and double vortex generators on the S809 airfoil at low Reynolds numbers, employing a combination of experimental and numerical techniques to identify the best positions for VGs to improve aerodynamic performance. The results indicated enhancements in lift force, delayed stall occurrence, and increased flow energization in the wake region. Serdar Genç, M. et al. [22] applied sandpaper to the airfoil and concluded that it offers significant advantages by delaying stall and improving the aerodynamic performance of wind turbine blades. Serdar Genç, M. et al. [23] explored applying partially flexible membrane material to both surfaces of the NACA 4412 airfoil and determined that its addition effectively controlled flow separation, enhancing aerodynamic performance at low Reynolds numbers.

From studying the above literature survey, this study looks into how adding a slot and groove to the NACA 4412 airfoil might improve its aerodynamic performance. By using numerical simulations and experiments, the authors aim to understand how these modifications affect the airfoil's ability to handle airflow. They are particularly interested in seeing if these changes can help the airfoil maintain lift at higher angles of attack and delay airflow separation. For this purpose, a L shaped groove is used at 50 % of the chord length and 3.5 mm depth at the end of the slot, comparing its performance with the slot-only configuration. Therefore, this research provides a comprehensive investigation into the potential of slot and groove modifications, showcasing their significance for aeronautics and the advancement of aerodynamic performance.

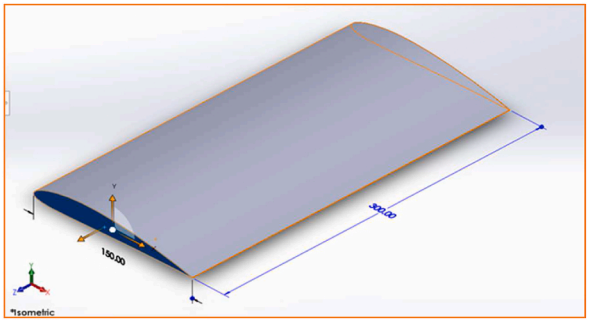
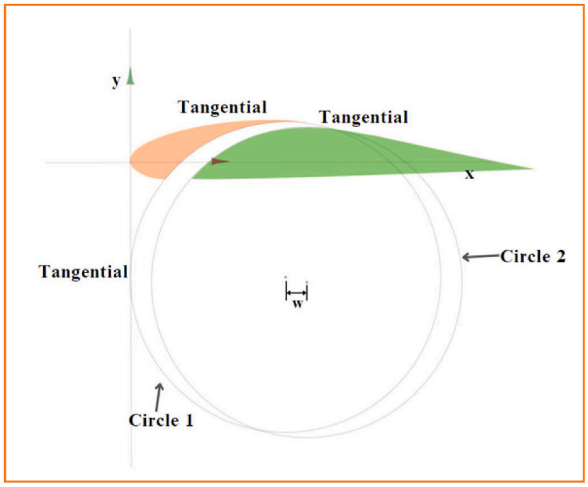
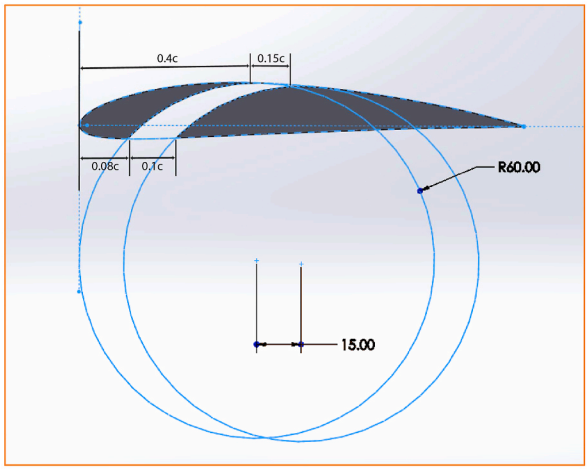


Fig. 1. Normal NACA 4412 airfoil.



(a)



(b)

Fig. 2. (a) Geometric pattern for internal slot (b) slotted NACA 4412 airfoil with dimensions.

2. Design method

The NACA 4412 airfoil is selected for this experiment due to its characteristics of low lift coefficients and relatively high drag coefficients [24]. Initially, wing model is created in SOLIDWORKS to pursue this objective. Subsequently, these designs are translated into wooden wing models. Finally, the aerodynamic characteristics of these models are computed.

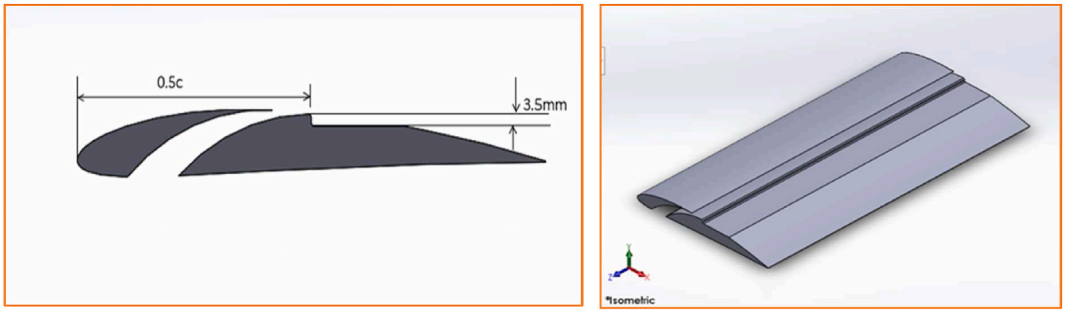


Fig. 3. NACA 4412 airfoil with internal slot and groove (a) side view, (b) isometric view.

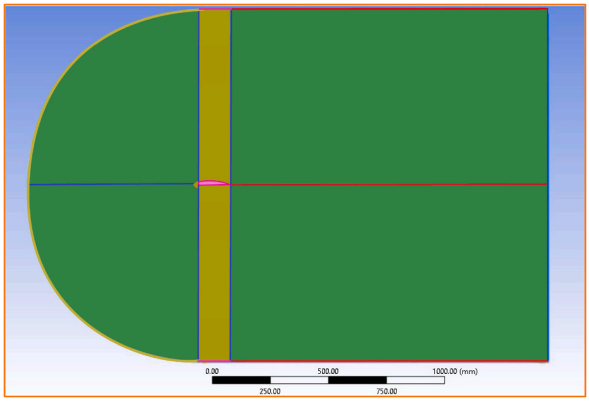


Fig. 4. Meshing methodology.

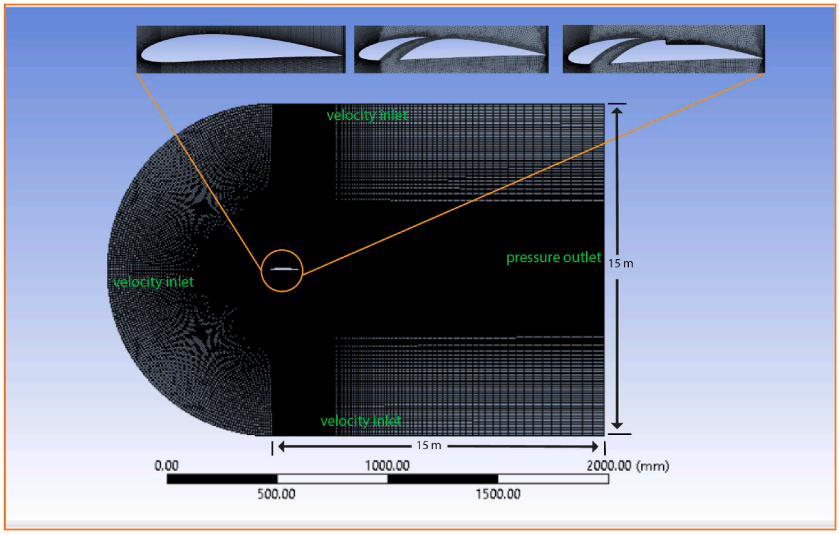


Fig. 5. Computational domain with boundary conditions.

2.1. Wing model

Using NACA 4412 airfoil, wooden models for three wings are prepared to have an equal surface area (45,000 mm^2), as shown in Fig. 1. For the wing, the chord length at the root and the span length are 150 mm and 300 mm, respectively.

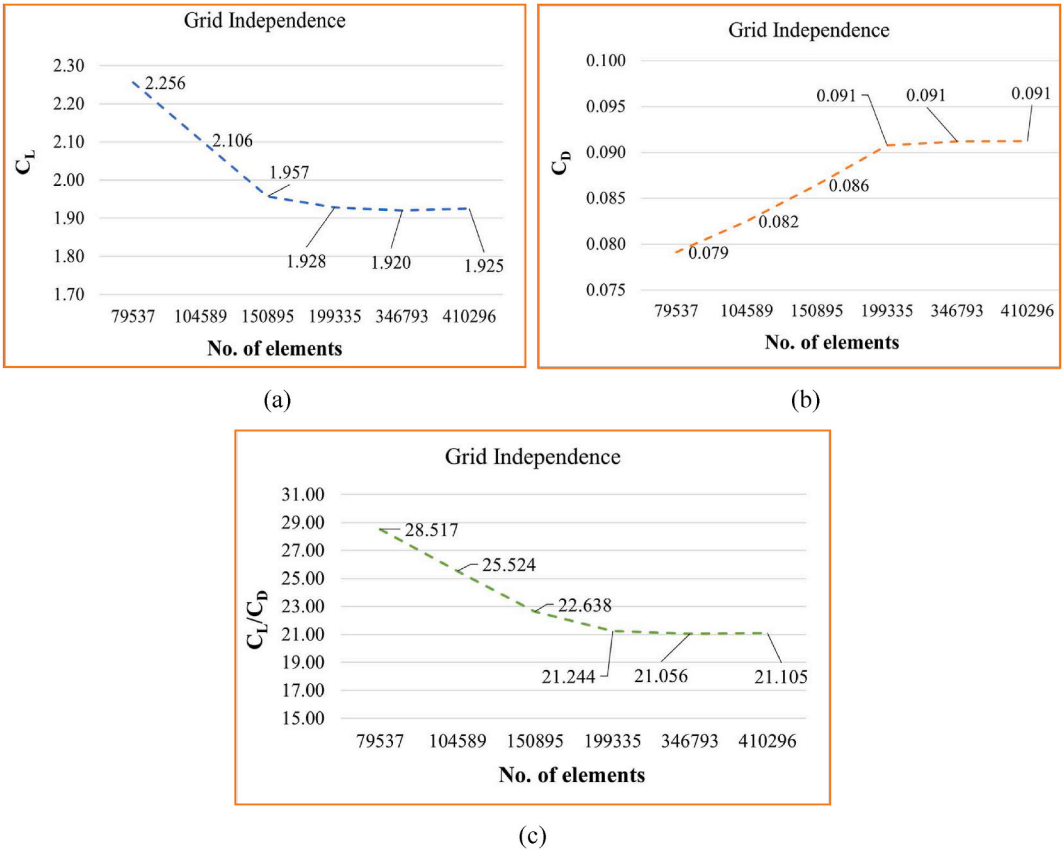


Fig. 6. Grid independence test of (a) lift coefficient, (b) drag coefficient, and (c) lift-to-drag ratio for different numbers of elements.



Fig. 7. Small components of the airfoil model.

2.2. Internal slot

An internal slot is utilized in order to create a pathway for high-momentum fluid to flow from the pressure surface to the suction surface, thereby preventing flow separation on the suction surface and enhancing aerodynamic performance. As shown in Fig. 2 (a), two circles (Circle 1 and Circle 2) of the same radius "r" are used to create the internal slot on the airfoil profile. Circle 1 is tangential to both the y-axis and the suction surface, whereas Circle 2 is only tangential to the suction surface. To properly steer the flow, circles must be tangent to the suction surface. The internal slot's width, "w," is the distance between both circle centers. Width "w" and radius "r" are the two key parameters that influence the effectiveness of the internal slot. The width is 0.1c, and the radius is 0.4c in the form of "w_r." Therefore, for our experiment, the internal slot 0.1c_0.4c has been chosen [6]. So, the values of w and r are 15 mm and 60 mm, respectively. In Fig. 2 (b), the slot modification in the NACA 4412 airfoil is shown with dimensions.



(a)

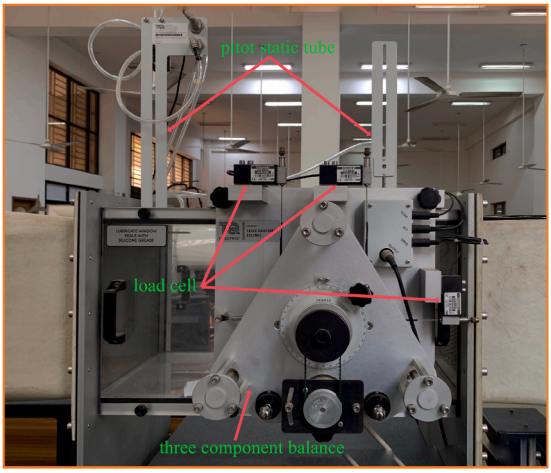


(b)

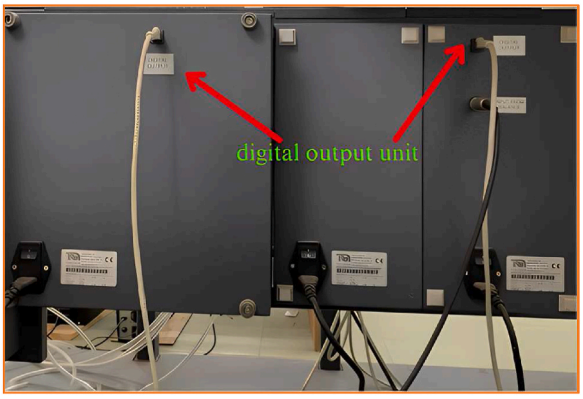


(c)

Fig. 8. Airfoil models for experiment: (a) normal NACA 4412 model, (b) NACA 4412 model with an internal slot, (c) NACA 4412 model with internal slot and groove.



(a)



(b)



(c)

Fig. 9. (a) Three-component balance (b) digital output unit (c) effuse.

2.3. Internal slot with groove

A groove can delay the boundary layer separation, which is vital in decreasing drag force. Thus, a groove can improve aerodynamic performance. A groove is inserted with a slot at 50 % of the chord length and a depth of 3.5 mm. Fig. 3 shows NACA 4412 airfoils with an internal slot and a groove.

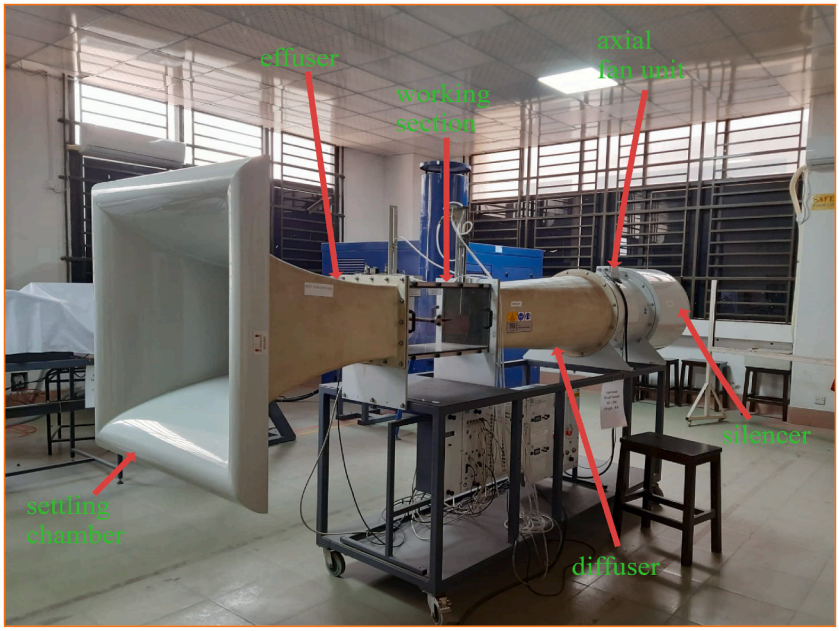
3. Computational analysis

The calculations are performed to analyze the aerodynamic characteristics at various AOAs, specifically within the range of 22°. The computational fluid dynamics of ANSYS is used to determine the lift and drag coefficients of the following airfoil models. Calibration of the turbulence model was not conducted explicitly in this study. Instead, the authors utilized the Spalart-Allmaras model, previously calibrated by a team of researchers [25] at the Bangladesh University of Engineering and Technology (BUET). This pre-established calibration was considered adequate, eliminating the need for additional adjustments in our analysis.

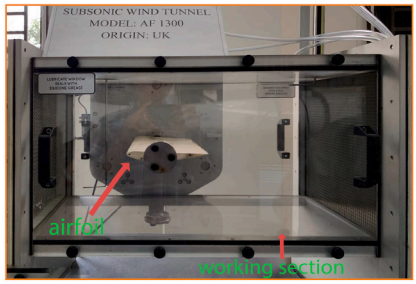
3.1. Design and methodology

According to the prospective design modification obtained from literature, airfoils are modeled in SOLIDWORKS. A C-type grid is chosen for the environment, and airfoils are designed as blank spaces on the environment surface. In ANSYS fluent, the design modeler is used to import those 2D-designed surfaces created in SOLIDWORKS and to add material as fluid.

The selection of the C-type grid for computational analysis necessitates a complex generation process to attain optimal accuracy. Consequently, meticulous testing is conducted to determine the ideal number of nodes and elements for the computational analysis while mitigating grid dependence. The surface body is partitioned into five distinct faces: two rectangular faces positioned behind the



(a)



(b)

Fig. 10. (a) VDAS AF1300 subsonic wind tunnel, (b) working section.

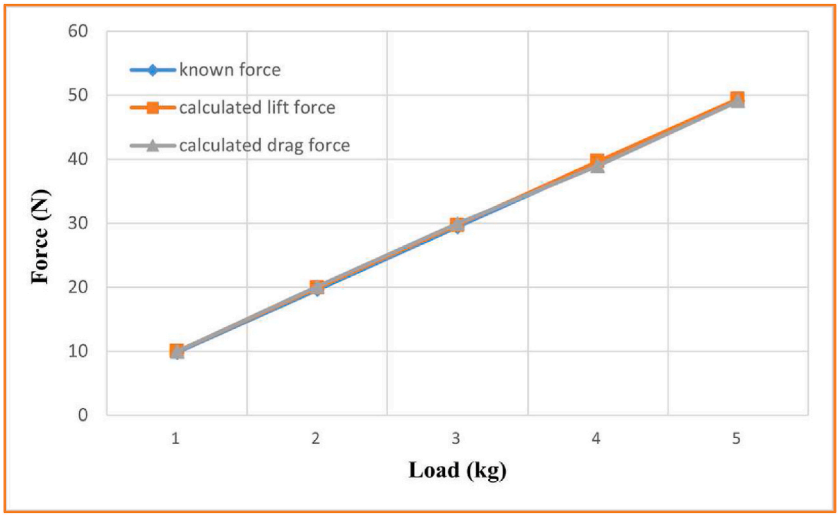


Fig. 11. Calibration of the wind tunnel.

airfoil's trailing edge, two quarter circles situated just ahead of the airfoil adjacent to the leading edge, and one rectangular face encompassing nearly the entirety of the airfoil's surroundings.

Mesh generation involves the utilization of three distinct meshing techniques: Edge Sizing, Face Sizing, and Face Meshing. Initially, seven edges (highlighted in blue) are selected, with 250 divisions and a bias factor of 100 employed to achieve a soft behavior. Subsequently, three edges (highlighted in red) are chosen for second-edge sizing, assigned 150 divisions and a bias factor of 300 to ensure a harder behavior. For the third edge sizing (highlighted in pink), 150 divisions without bias are selected to maintain a hard behavior. Finally, the fourth edge sizing (highlighted in yellow) employs 100 divisions without bias for a harder behavior. The four green faces are meshed using a quadrilateral mapping technique. The yellow region is particularly important as it contains most of the airfoil's components and modifications. We use a soft element sizing method here, with elements sized at 2 mm. Applying this methodology yields the following statistics: The NACA 4412 airfoil features 201,202 nodes and 200,300 elements. Introducing a slot to the NACA 4412 airfoil results in 192,943 nodes and 192,245 elements. NACA 4412 airfoil, including an internal slot and a groove, produce a configuration with 200,275 nodes and 199,335 elements. All the edges and airfoils are named according to the requirements of simulation as shown in Fig. 4.

3.2. CFD parameters and setup

For turbulence modelling, the Spalart-Allmaras model is used because of its several benefits, such as being relatively simple to implement and computationally efficient as it is a one-equation model, making it well-suited for our experiment. 2D models of a standard NACA 4412, a modified NACA 4412 with an internal slot, and a modified NACA 4412 with an internal slot and groove are designed in SOLIDWORKS. The coordinates for the models are obtained from the website www.airfoiltools.com and imported into SOLIDWORKS to create 2D airfoil models. Then wing models are imported into the ANSYS 2D CFD module. Fig. 5 shows the computational domain with boundary conditions.

For the boundary conditions, the inlet velocities of air are taken as 20 m/s and 43.9 m/s, respectively, and the pressure and temperature are taken as SATP conditions. The Reynolds numbers for 43.9 m/s and 20 m/s are 4.2×10^5 and 1.9×10^5, respectively. The dynamic viscosity is 1.849×10^{-5} kg/m-s and the density is taken as 1.184 kg/m^3. Turbulent intensity and viscosity ratio conditions are set to a value of 1. The gauge pressure is 0. At the outlet and the wall, the ambient atmospheric conditions and no-slip boundary conditions are applied. The upper and lower surfaces of the airfoil are treated as wall boundaries.

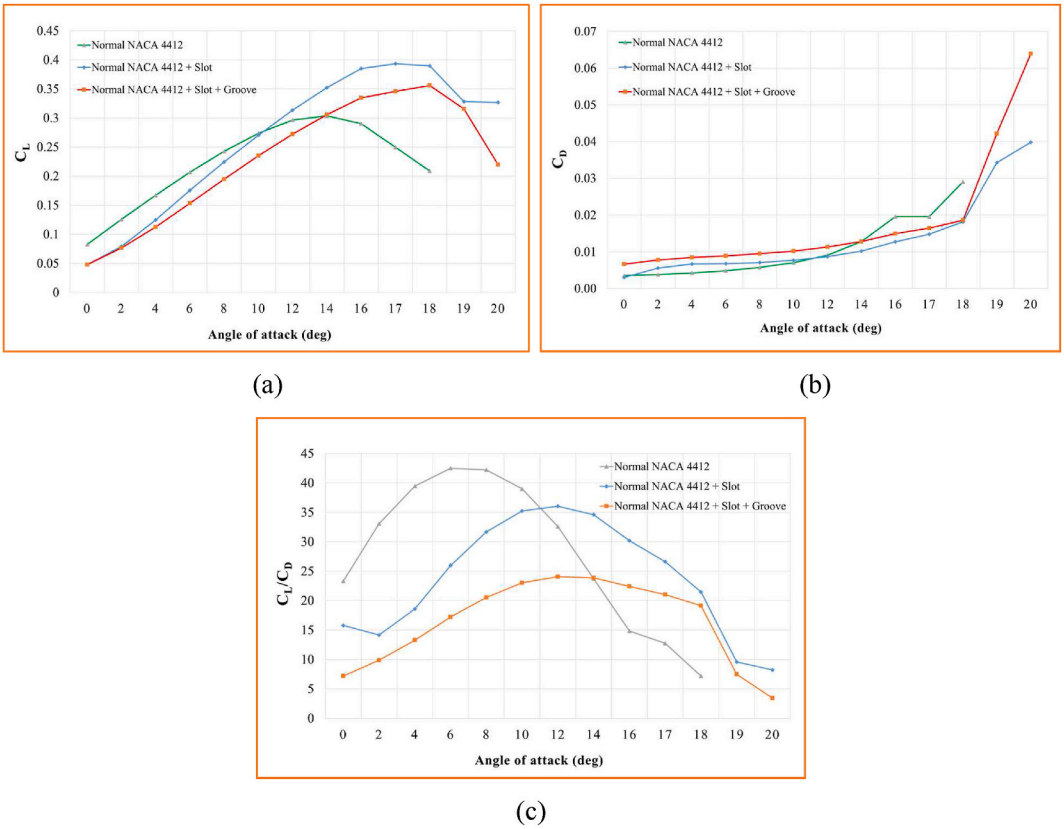


Fig. 12. Comparison of simulated data for (a) lift coefficient, (b) drag coefficient, and (c) lift-to-drag ratio at 20 m/s.

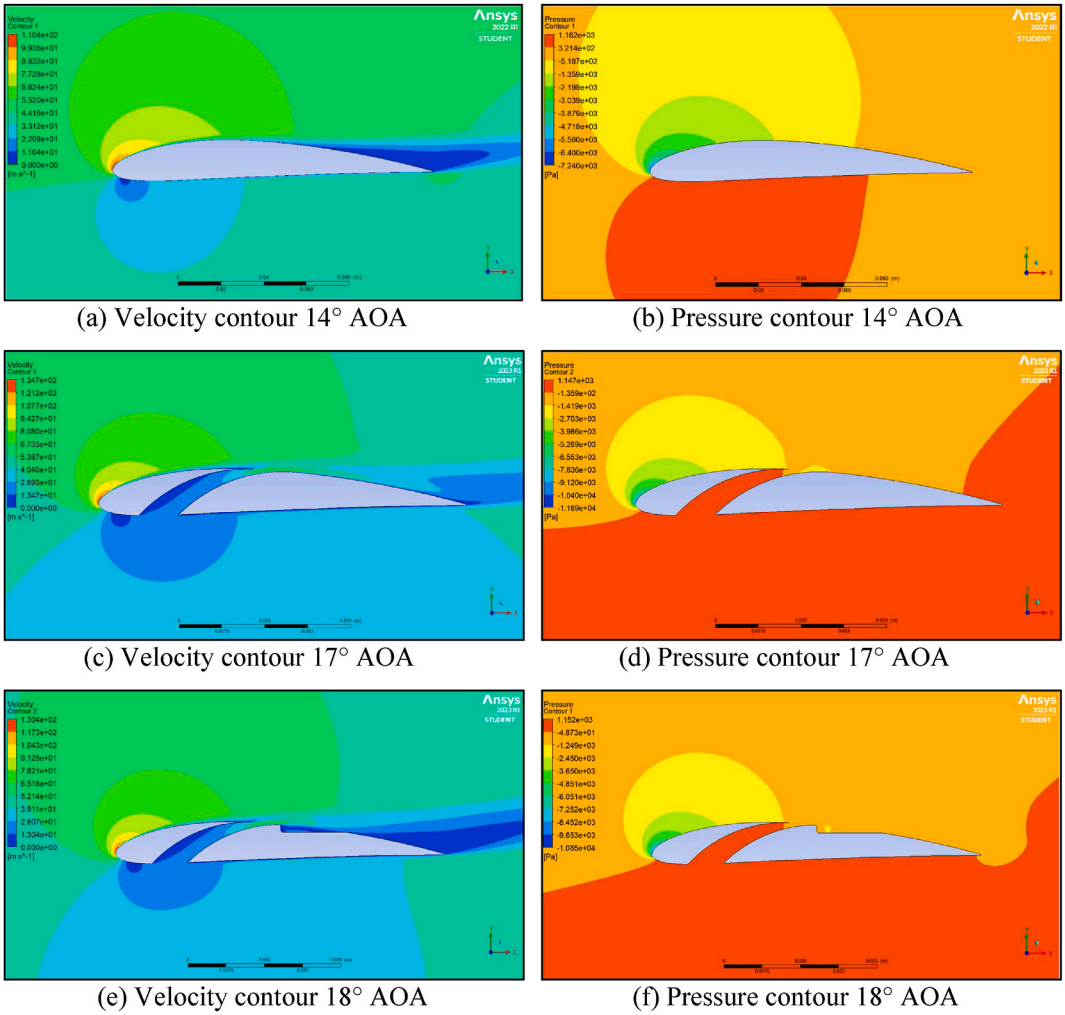


Fig. 13. Velocity and pressure contours of three airfoils at their stall angle at 20 m/s.

3.3. Grid independence

To ensure precision of numerical analysis, an optimal grid design is essential, typically assessed through a grid independence test. Grid shape, quality, and quantity are critical considerations in crafting effective grids [26]. Fig. 6 shows the grid independence study on a NACA 4412 airfoil with slot and groove at 20° AOA at a free stream velocity of 43.9 m/s, and the lift coefficient (C_L), drag coefficient (C_D), and lift-to-drag ratio (C_L/C_D) obtained from the results are plotted in the figure.

4. Experimental analysis

4.1. Airfoil fabrication

In this experiment, three airfoils are fabricated from balsa wood sheets with a thickness of 5 mm. The initial step involves using laser cutting technology to create smaller components, as illustrated in Fig. 7. A total of 60 segments, each 5 mm in thickness, are employed to construct the assembly, resulting in a chord and span length of 150 mm and 300 mm, respectively. During this assembly process, thin wood adhesive is applied to bond the pieces together, thus forming the three-dimensional configuration of the airfoil. To enhance the airfoil's rigidity and structural integrity, three support rods, each 2 mm in diameter, are strategically inserted through its span.

After the assembly is completed, any excess glue is carefully removed by pressing the airfoils along its chord length. The surface of the airfoil is then smoothed using a bench grinder machine and fine grit sandpaper in sequence. To prepare the airfoil for wind tunnel testing, a hole with a diameter of 10 mm $\pm$ 0.2 mm is drilled using a milling machine, and a stainless-steel rod is inserted into this hole, as depicted in Fig. 8.

4.2. Wind tunnel

The experiment is conducted using the VDAS AF1300 low-speed subsonic wind tunnel, as depicted in Fig. 10(a). This open-circuit wind tunnel features a test section measuring 305 mm by 305 mm. The working section of the tunnel is a square with transparent ceiling, sides, and floor, allowing for easy observation. The sides are removable for convenience. The wind tunnel is equipped with a protractor and a model holder to support and adjust the angle of any installed models. Measurements such as pitching moment can be taken using a three-component balance, illustrated in Fig. 9 (a). This instrument uses high-quality load cells connected by wires. When an airfoil exerts lift or drag force, these wires stretch, causing a small voltage in the load cell. This voltage is then amplified and converted into actual force. The data acquisition system processes the signal from the load cells and converts the raw force data into lift and drag coefficients.

Two pitot-static tubes located on the top of the working section are used to measure the mean velocity. These tubes can determine the velocity of the flow by comparing the static pressure and the total pressure. Two wall tappings on the top of the working section measure the static pressure upstream and downstream. Air enters the tunnel through an effuser (contraction cone) that linearly accelerates the air. It then passes through a honeycomb flow straightener and enters the working section. The air subsequently passes through a grill, a diffuser, and finally a variable speed axial fan. The grill protects the fan from stray items. The air exits the fan, travels through a silencer, and then returns to the atmosphere. The axial fan's speed is controlled by a separate control and instrumentation unit. The wind tunnel's instrumentation includes electronic sensors that can be connected to Versatile Data Acquisition System (VDAS). When connected to a suitable computer, VDAS provides accurate real-time data collection, monitoring, display, and calculation of all relevant parameters.

The wing model has been placed in the working section of the wind tunnel, as depicted in Fig. 10(b). All experimental data are collected at 25 °C, and the airflow is considered steady and incompressible throughout the experiment. The wing models are tested at an angle of attack from 0 to 21° with an interval of two degrees. The values of lift and drag coefficients are found for an angle of attack of 0°, 2°, 4°, 6°, 8°, 10°, 12°, 14°, 16°, 18°, 20°, and 21°, respectively, at a velocity of 20 m/s for all three airfoils. The Reynolds number are taken as 1.9×10^5.

4.3. Calibration of the wind tunnel

The calibration process of the VDAS AF 1300 wind turbine, used in this research, begins by setting up the three-component balance

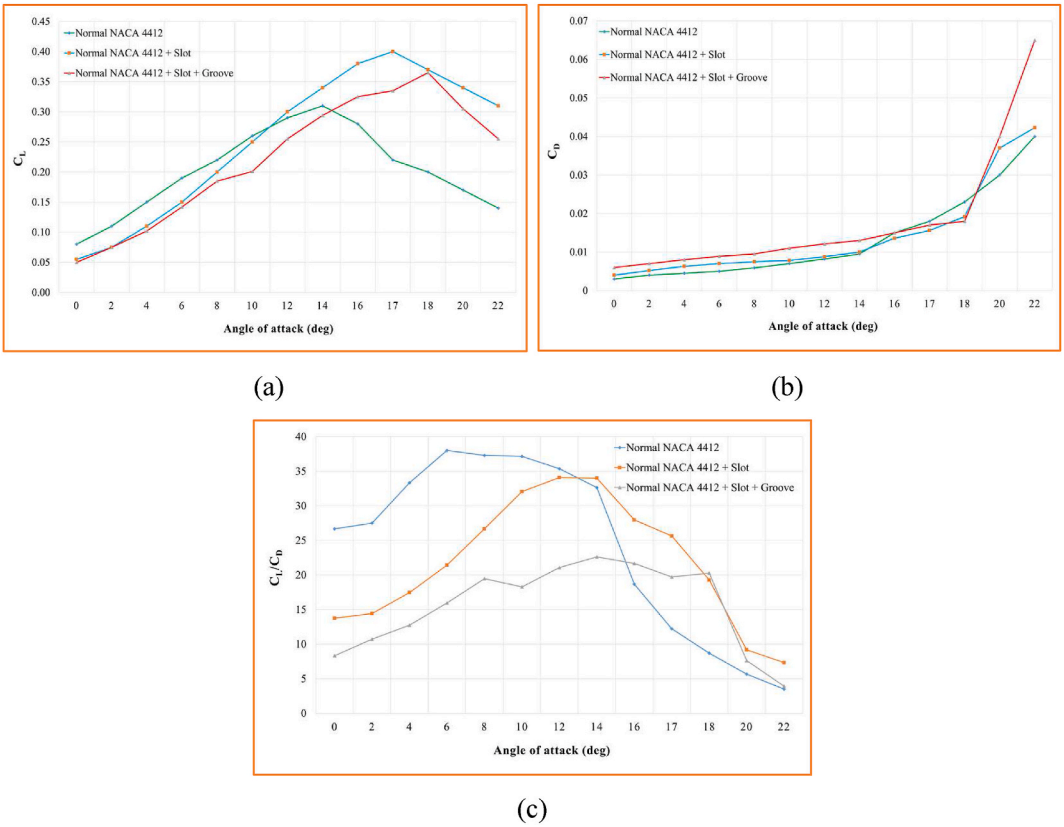


Fig. 14. Comparison of experimental data for (a) lift coefficient, and (b) drag coefficient, and (c) lift-to-drag ratio at 20 m/s.

on a stand and leveling it in three dimensions using a straight level. This ensures that forces applied during experimentation are evenly distributed across the load cells, minimizing inaccuracies. A load hanger is then attached to hang the load. Lift and drag are calculated separately, with values obtained from a digital pressure transducer. Lift is measured by applying the load vertically, while drag is measured by hanging the load horizontally using an attachment. Different types of loading are tested, ensuring the load hangs freely over the edge of the table, with a maximum load of up to 5 kg. The percentage of error is calculated for each known loading or weight, with a maximum acceptable error of up to 5 %. If the error lies within this range, the wind tunnel is deemed ready for use. If the error exceeds this range, adjustments are made to the wire in the load cell and the procedure is repeated. This process ensures the wind turbine's performance measurements are accurate and reliable (see Fig. 11).

5. Results and discussion

In this research work, the numerical investigations are conducted at two distinct velocities, namely 20 m/s and 43.9 m/s, while the experimental analysis is executed at 20 m/s due to the limitations and safety measurements of the subsonic wind tunnel. All the data, the graph, and its validation are given below.

5.1. Aerodynamic performance at lower velocity

5.1.1. Numerical study

From the numerical study, it is found that at the velocity of 20 m/s, the stall angles for NACA 4412, NACA 4412 with a slot, and NACA 4412 with a slot and a groove are 14°, 17°, and 18°, respectively, as shown in Fig. 12. The graph clearly shows that, among the three airfoils, the NACA 4412 equipped with a slot demonstrates the superior lift coefficient. The numerical results are highlighted in the accompanying graph, which shows that, beyond these angles, the lift coefficient noticeably decreases and the drag coefficient increases.

To demonstrate the effects of airspeed, pressure, and flow separation around the airfoils, velocity and pressure contours are depicted in Fig. 13. These contours illustrate the aerodynamic properties of the region around the airfoil at a constant velocity of 20 m/s, corresponding to their respective stall angles of 14°, 17°, and 18° for the NACA 4412, NACA 4412 with a slot, and NACA 4412 with both a slot and a groove. A significant amount of flow separation is clearly visible in the velocity contours, and it reaches approximately 50 % of the chord length for all three airfoils. In contrast, the pressure contour clearly indicates that the pressure on the lower surface of

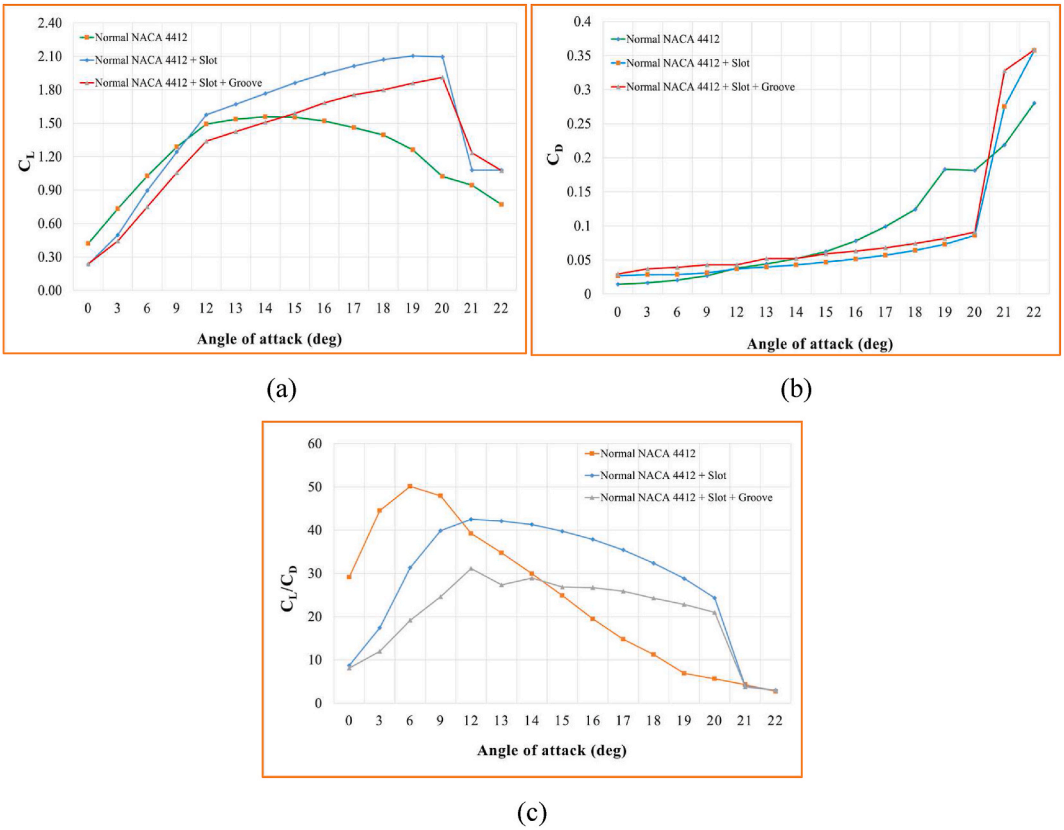


Fig. 15. Comparison of simulated data for (a) lift coefficient, (b) drag coefficient, and (c) lift-to-drag ratio at 43.9 m/s.

the airfoil increases from the normal configuration to the airfoil with a slot and groove until it reaches the stall angle.

5.1.2. Experimental study

The experimental study is done only at 20 m/s considering the safety measurements of the wind tunnel. The generated lift coefficient, drag coefficient, and lift-to-drag ratio from the experiments are shown in Fig. 14. The comparative analysis of numerical and experimental data highlights distinctive stall angles for three variations of the NACA 4412 airfoil: normal NACA 4412, NACA 4412 with a slot, and NACA 4412 with a slot and a groove. The measured stall angles are 14°, 17°, and 18°, respectively. Notably, the NACA 4412 with a slot demonstrates the highest lift generation at higher AOAs, exhibiting an optimized stall angle of 17°. In contrast, the normal NACA 4412 airfoil initially produces the highest lift but experiences lower lift forces at higher AOAs. The NACA 4412 with a slot and groove outperforms the others by achieving the highest stall angle of 18°, although it exhibits comparatively lower lift forces at lower AOAs.

5.2. Aerodynamic performance at higher velocity

For the numerical study at higher velocity, airfoils are fixed against a velocity of 43.9 m/s. In comparison to the normal airfoil, the modified airfoils exhibit a decrease in drag coefficient and an increase in lift coefficient, as illustrated in Fig. 15. Notably, at higher velocity, the modified airfoils demonstrate a nearly 2° increase in stall angle compared to their performance at lower velocity. The stall angles at a velocity of 43.9 m/s for NACA 4412, NACA 4412 with slot, and NACA 4412 with slot and groove are 14°, 19°, and 20°, respectively.

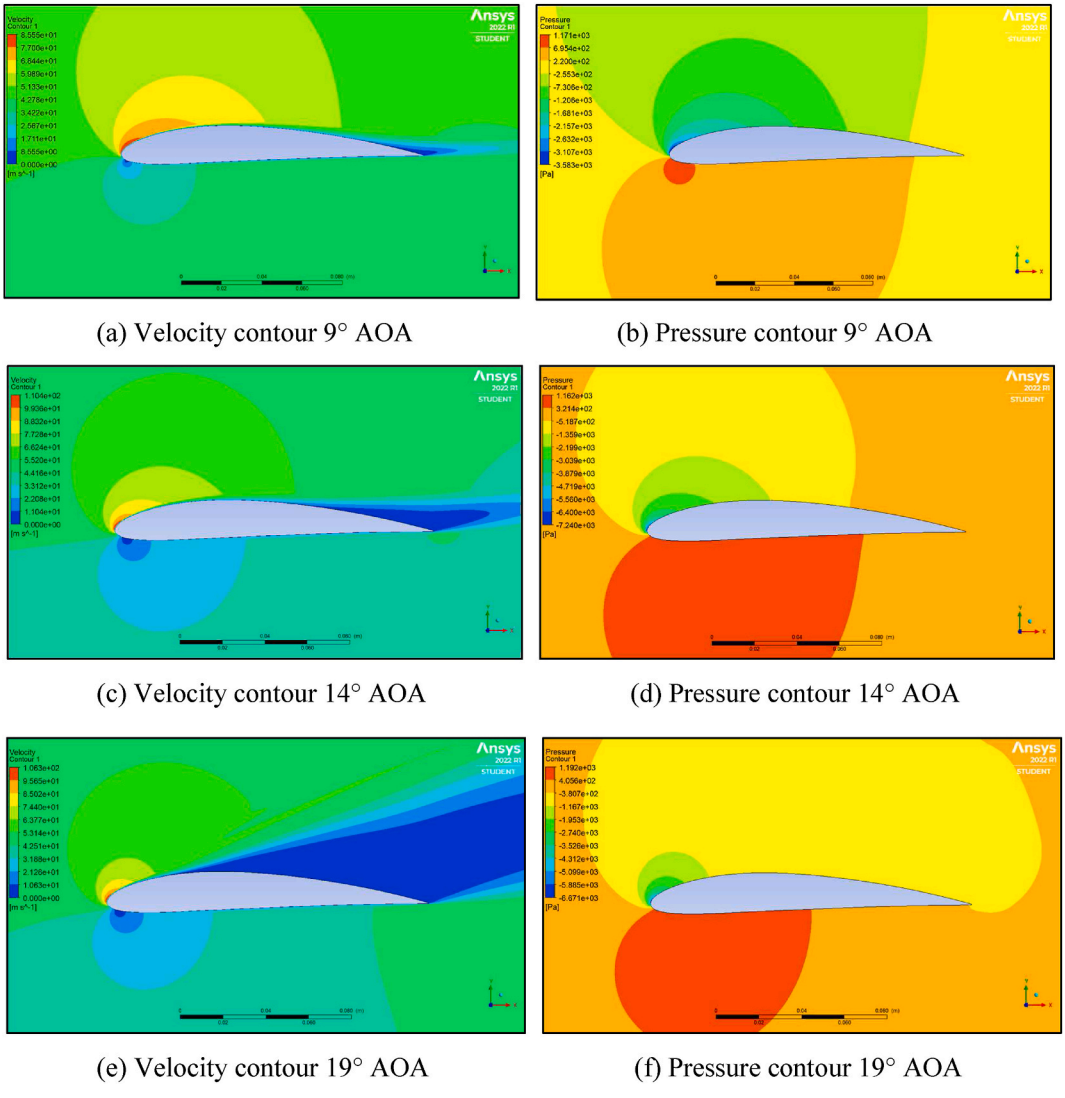


Fig. 16. Velocity and pressure contours of normal NACA 4412 airfoil at 43.9 m/s.

The velocity and pressure contours of the standard NACA 4412 airfoil are shown in Fig. 16, where the contours are represented at an angle of 9°, 14°, and 19°. These AOAs are chosen to see the change(s) in velocity and pressure contours of normal airfoil in the pre-stall, stall, and post-stall cases. The velocity contour analysis reveals a consistent pattern of flow separation extending from the trailing edge to the leading edge of the airfoil with an increase in the angle of attack. At an AOA of 9°, flow separation is observed at the edge of the trailing edge. However, at 14° AOA, flow separation moves closer to the leading edge, occurring at approximately 60 % of the chord length. After stall, such as at 19° AOA, flow separation occurs at approximately 25 % of the chord length. On the other hand, the pressure contour demonstrates a gradual increase in pressure on the lower surface of the airfoil, corresponding to the angle of attack, until it reaches the stall angle of 14°. Beyond this point, there is a subsequent reduction in the lift coefficient as the AOA continues to increase. From the pressure contour color legend, it is evident that at lower AOAs, the pressure on the lower surface of the airfoil is higher than that on the upper surface. As the AOA increases, both upper and lower surface pressures increase. However, when the airfoil exceeds its stall angle, the highest pressure is observed on the lower surface, but it is significantly less than the pressure on the upper surface, as depicted in Fig. 16(f). Consequently, at this AOA, the airfoil is unable to generate any lift force.

In Fig. 17, the aerodynamic characteristics of the NACA 4412 airfoil with a slot are presented. The slot acts as a passageway for incoming air, causing a notable shift in flow separation towards the airfoil's trailing edge compared to the normal NACA 4412 airfoil, as evidenced by the velocity contours. Fig. 17(a), (c) and 17(e) reveal that air passes through the slot region, and a stagnation point is also seen within the slot structure. The airflow through the slot helps in reducing the velocity gradient on the upper surface of the airfoil, thereby decreasing the boundary layer effect and delaying flow separation. As a result, drag force is reduced at high AOAs compared to a normal airfoil. Therefore, it can be concluded that incorporating a slot can effectively delay flow separation and can produce lift at high AOAs. By observing the pressure contour legend in Figs. 16(b) and 17(b), we can conclude that the lift force of the

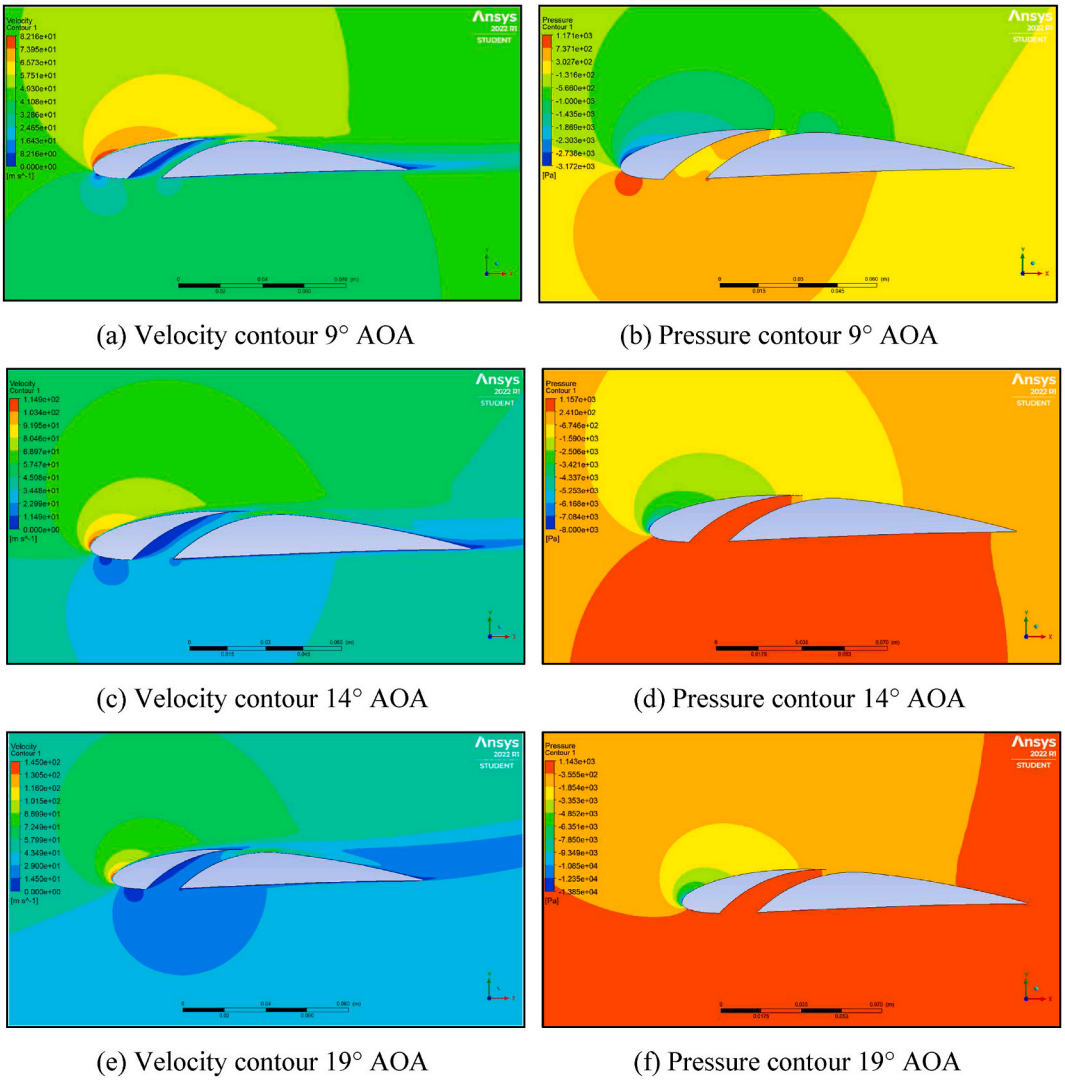


Fig. 17. Velocity and pressure contours of NACA 4412 airfoil with slot at 43.9 m/s.

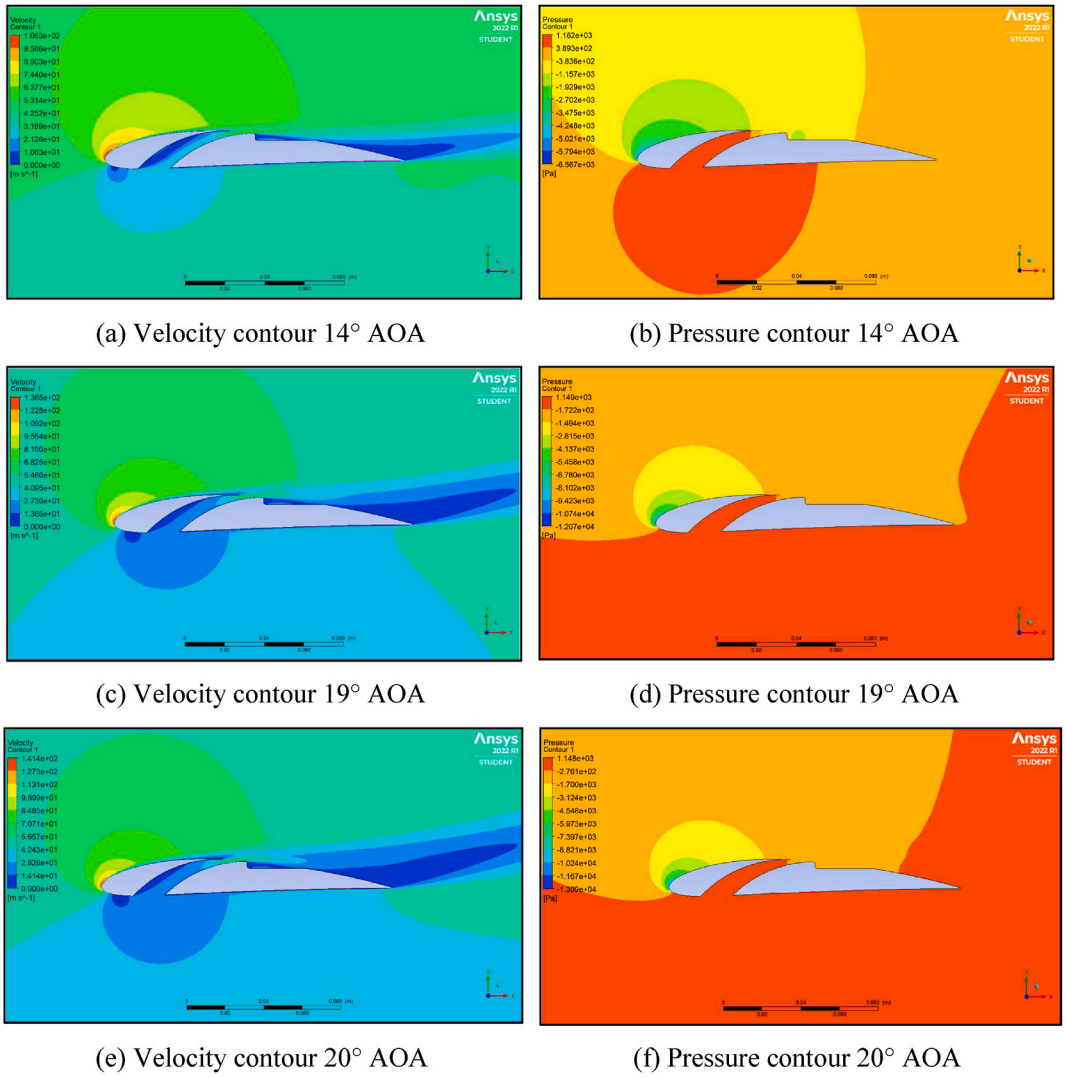


Fig. 18. Velocity and pressure contours of NACA 4412 airfoil with slot and groove at 43.9 m/s.

slotted airfoil is lower than that of the normal NACA 4412 at an AOA of 9°. This is due to some air passing through the slot, which contributes to lift in the normal airfoil. However, beyond a 12° AOA, the lift force of the slotted airfoil exceeds that of the normal NACA 4412 as the pressure in the slot region starts to increase, thus enhancing the lift force in the slotted airfoil. Comparing Figs. 16(d) and 17(d), we can see that for the same AOA (14°), the pressure on the lower surface of the slotted airfoil is much higher than the normal airfoil. Thus, the lift force of the slotted airfoil at 14° AOA is much higher than normal airfoil. After the stall angle, an increase in pressure on the upper surface of the airfoil is observed, resulting in a decrease in lift force, as shown in Fig. 17(f).

In Fig. 18, the velocity and pressure contours of NACA 4412 with a slot and groove are presented. Flow separation normally occurs at the trailing edge at lower velocities but shifts towards the leading edge as the AOA increases. From the analysis of the contours of the slotted airfoil, it can be said that the slot structure helps to reduce the velocity gradient. This can help to minimize the formation of boundary layers and delay flow separation. Moreover, incorporating a groove immediately after the slot further moves the flow separation towards the trailing edge, effectively increasing the stall angle, as illustrated in Fig. 18. As some airflow passes through the slot and then the groove structure, it prevents the mainstream flow from touching the surface of the airfoil. Hence, it helps to overcome the boundary layer effect. Therefore, the combined use of a slot and groove further delays the flow separation towards the trailing edge. Comparing Figs. 17(f) and 18(d), we observe that at the same angle of attack (19°), the pressure on the upper surface of the slotted airfoil is significantly higher than that on the slot and grooved airfoil. Consequently, at 19° AOA, the slotted airfoil is unable to produce any lift, whereas the slot and grooved airfoil can still generate some amount of lift. At 20° AOA, as shown in Fig. 18(f), an increase in pressure on the upper surface of the airfoil is observed, resulting in a decrease in lift force. Beyond this angle (20°), the airfoil is unable to generate any lift force.

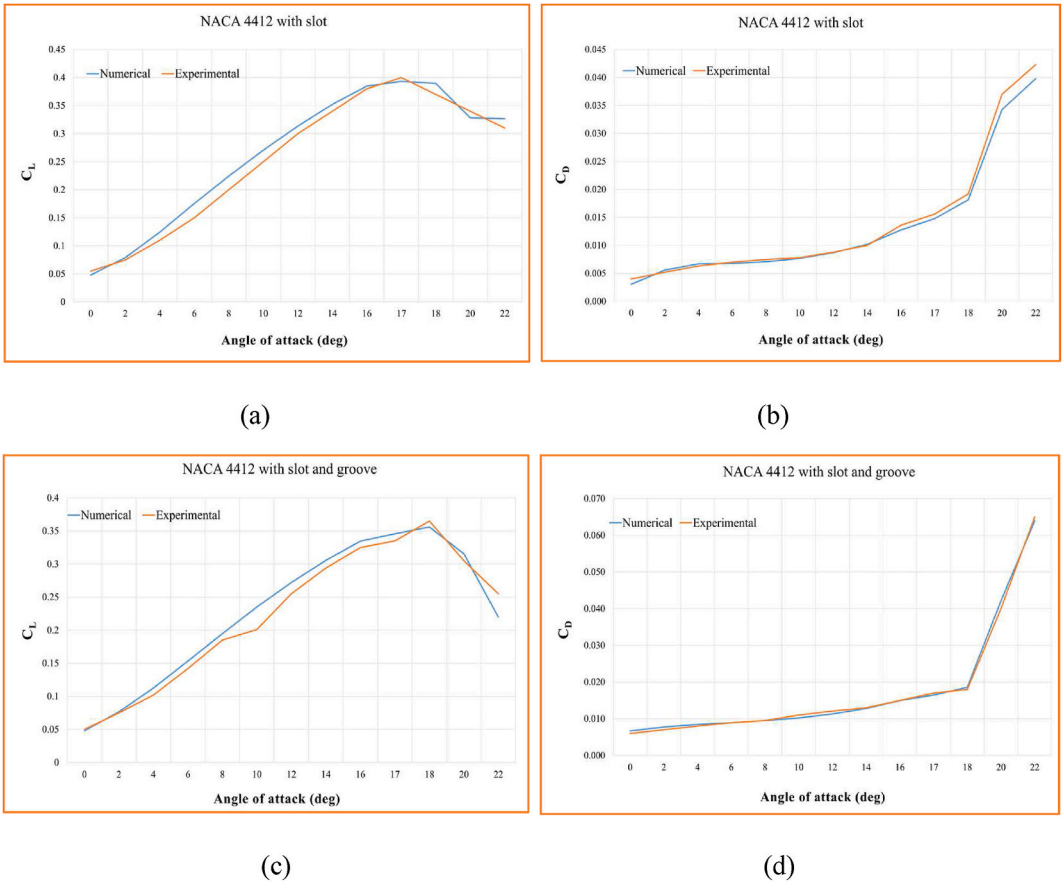


Fig. 19. Comparison between experimental and numerical data at 20 m/s of (a) lift coefficient, (b) drag coefficient for NACA 4412 with slot and (c) lift coefficient, (d) drag coefficient for NACA 4412 with slot and groove.

5.3. Validation

At a velocity of 20 m/s, both experimental studies and numerical simulations are conducted, thus establishing this velocity as the benchmark for validating our research findings. For validation purposes, both modified airfoils, namely the NACA 4412 airfoil with a slot and the NACA 4412 airfoil with both a slot and a groove, are utilized in this research. A comparative analysis of the experimental data, comprising lift and drag coefficients, and the corresponding numerical data, is presented in Fig. 19. Discrepancies in the observed values may be attributed to various factors, including environmental conditions, fluctuations in air speed, and fabrication errors in airfoils. Nevertheless, the experimental data corresponds to the simulated data, indicating consistency in the observed trends.

5.4. Variation of stall angle with wind velocity

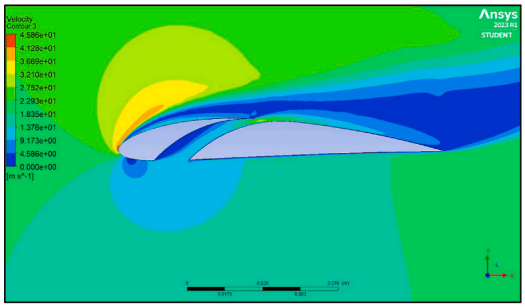
The data analysis from the CFD module indicates that when the incoming air velocity is higher, the modified airfoils stall angle increases accordingly. To visually compare the velocity contour differences at the same AOA, we can compare Fig. 20(a) and Fig. 17(e).

Fig. 20(a) displays the velocity contour for a slotted airfoil at an AOA of 19° and an airspeed of 20 m/s. On the other hand, Fig. 17(e) represents the velocity contour for the same slotted airfoil, also at an AOA of 19°, but at a higher speed of 43.9 m/s. This comparison allows us to observe the changes in the velocity contour of a slotted airfoil at the same AOA but different speeds.

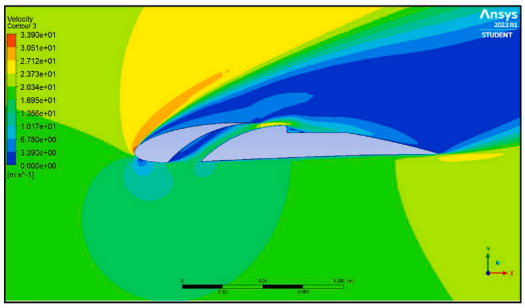
As seen in Fig. 20(a), at a lower airspeed of 20 m/s, the flow separation point starts at the leading edge. The velocity gradient is much higher in the case of lower velocity, as shown in Fig. 20(a). However, at a higher speed of 43.9 m/s, air passes through the slot region at high velocity, moving the flow separation point closer to the trailing edge from the leading edge. Because of the higher velocity of the air coming through the slot region, the velocity gradient decreases, which is evident in Fig. 17(e). As the velocity gradient decreases for higher velocity, it prevents the formation of the boundary layer. Therefore, by delaying the boundary layer formation, the flow separation is also delayed, which means the stall angle can increase. Thus, the stall angle increases in the case of higher velocity. That is why, in the slotted airfoil, the stall angle is 19° at higher velocity and 17° at lower velocity.

Similar study can be done for NACA 4412 airfoil with slot and groove. We can compare Fig. 20(b) with Fig. 18(e).

Fig. 20(b) displays the velocity contour for NACA 4412 airfoil with slot and groove at an AOA of 20° and an airspeed of 20 m/s. On



(a) NACA 4412 with slot at an AOA of 19° at 20 m/s



(b) NACA 4412 with slot and groove at an AOA of 20° at 20 m/s

Fig. 20. Velocity contours at lower velocity.

the other hand, Fig. 18(e) represents the velocity contour for the same slot and groove airfoil, also at an AOA of 20°, but at a higher speed of 43.9 m/s. This comparison allows us to observe the changes in the velocity contour of a slot and grooved airfoil at the same AOA but different speeds.

As seen in Fig. 20(b), at a lower airspeed of 20 m/s, the velocity gradient is much higher as compared to the velocity gradient seen in Fig. 18(e). Because of the higher velocity airflow passes through the slot and then the groove structure, it prevents the mainstream flow from touching the surface of the airfoil. Hence, it helps to overcome the boundary layer effect more than the case in low velocity. Therefore, the velocity gradient decreases more, thus the flow separation delays future more in the case of higher velocity than the case in low velocity. As a result, in the NACA 4412 airfoil with slot and groove, the stall angle is found to be 20° at higher velocity and 18° at lower velocity. This illustrates how the groove's effect vary with airspeed, influencing the stall characteristics of the airfoil.

6. Conclusion

In this study, the aerodynamic performance of the NACA 4412 airfoil, incorporating a slot and a slot with a groove, has been analyzed and compared. For this study, an internal slot located at 10 % and 40 % of the chord length (0.1c and 0.4c, respectively) has been chosen. In the grooved airfoil, a groove is inserted with a slot at 50 % of the chord length and a depth of 3.5 mm. Numerical simulations are done at airspeeds of 20 m/s and 43.9 m/s, while experimental investigations are conducted at a speed of 20 m/s. Both numerical simulations, utilizing the Spalart-Allmaras turbulence model, and experimental investigations in the VDAS AF1300 subsonic wind tunnel have been conducted, yielding similar findings. Based on the analysis of the results, the following conclusion can be drawn.

- Incorporating a slot, as well as a slot with a groove, effectively delays flow separation.
- At an airspeed of 20 m/s, the stall angles are observed to be 14°, 17°, and 18° for the normal NACA 4412, NACA 4412 with a slot, and NACA 4412 with both a slot and groove, respectively.
- At an airspeed of 43.9 m/s, the stall angles are 14° for the normal NACA 4412, 19° for NACA 4412 with a slot, and 20° for NACA 4412 with both a slot and a groove.
- A small increment of 2° in stall angle is observed at higher velocities for the modified airfoils.
- The NACA 4412 airfoil with a slot exhibits an improvement in lift coefficient, approximately 35 % higher than the normal NACA 4412 at higher AOAs. Moreover, the NACA 4412 with both a slot and groove achieves an almost 16 % higher lift coefficient than the normal airfoil at higher AOAs.
- The modified airfoils exhibit less lift-to-drag ratio at low AOAs compared to the normal airfoil. But the lift-to-drag coefficient shows noticeable improvement at higher AOAs for the modified configurations.

Overall, this comprehensive investigation, with the detailed experimental setup, demonstrates the promising aerodynamic benefits

of incorporating a slot and groove on an asymmetric airfoil. These findings can contribute to advancing our understanding of airfoil performance.

Data availability

Data included in article/supp. material/referenced in article.

Additional information

No additional information is available for this paper.

CRediT authorship contribution statement

Asif Mahmud Rayhan: Writing – original draft, Validation, Software, Methodology, Investigation, Data curation, Conceptualization. **Md Shahriar Hossain:** Writing – review & editing, Writing – original draft, Visualization, Resources, Methodology, Investigation, Conceptualization. **Rokeiya Haque Mim:** Methodology, Investigation, Conceptualization. **Mohammad Ali:** Writing – review & editing, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Nomenclature

AOA	Angle of Attack
AOAs	Angles of Attack
CFD	Computational Fluid Dynamics
C_L:	Lift coefficient
C_D	Drag Coefficient
C_L/C_D	Lift-to-Drag Ratio
VDAS	Versatile Data Acquisition System

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